THE PULLBACK OF A SMALL ETALE SHEAF, DESCRIBED ON THE BIG ETALE SITE

THE STACKS PROJECT AUTHORS

Editorial note. The counted statement is the author's own lemma environment labelled lemma-describe-pullback (source lines 747-758); the proof environment beginning on the line after the following blank line (source line 760) is its own proof and closes at source line 797.

Editorial note. The proof invokes one result of the same chapter by label (lemma-surjective-proper); it is retained as a context span and the generator expands the context transitively, so that reference resolves inside the package.

Editorial note. Three cross-chapter results are cited (Topologies on Spaces twice and Etale Cohomology once); they are declared as tagged source pointers, which is what the project's pointer rule allows, and nothing they would have to prove is used as a step of the counted argument.

Editorial note. The author's text leaves two verifications to the reader in the fpqc part ('We omit the verification ...'); they are reproduced verbatim and the card supplies nothing in their place.

Editorial note. The retained preamble is the chapter's own (sources/stacks-spaces-more-cohomology/preamble.tex, taken with its real line count), a verbatim excerpt of the native Xy-pic setup and theorem declarations; no macro was invented or redefined.

Editorial scope. One original result of the Stacks Project's chapter Cohomology of Algebraic Spaces, counted once: the pullback of a small etale sheaf described on the big etale site, and its sheaf property for smooth, syntomic, fppf, fpqc and ph coverings.

Allowed standard prerequisites (editorial).

- The small and big etale sites and their projections. The projection pi_X from the big etale site of algebraic spaces over X to the small etale site X_etale, the comparison map a_X of topoi, and the identity f_small = pi_X o i_f for f : Y -> X.. It fixes the notation of the statement and is the identity used in the first step (source lines 749-755 and 761-763).
- The ph topology and its equalizer criterion. A presheaf on the big etale site is a sheaf for the ph topology exactly when every surjective proper morphism V -> U of algebraic spaces over X gives an equalizer diagram of its sections (Topologies on Spaces, declared as a tagged source pointer).. It reduces the ph part of the proof to the equalizer statement at source lines 767-774.

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- The author's lemma on proper coverings. For a surjective proper morphism V -> U of algebraic spaces over X the sections of pi_X^{-1}F form the equalizer of the two projections (the author's lemma-surjective-proper, retained as context).. It supplies the input of the ph reduction at source line 774.

Source reference. Original lemma 0DFM: sources/stacks-chow/tags.txt, source lines 16842–16842. Cross-chapter pointer whose target carries the canonical tag 0DFM in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 0DFH: sources/stacks-chow/tags.txt, source lines 16837–16837. Cross-chapter pointer whose target carries the canonical tag 0DFH in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 09XN: sources/stacks-chow/tags.txt, source lines 12536–12536. Cross-chapter pointer whose target carries the canonical tag 09XN in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 04K2: sources/stacks-chow/tags.txt, source lines 5958–5958. Cross-chapter pointer whose target carries the canonical tag 04K2 in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 089J: sources/stacks-chow/tags.txt, source lines 10499–10499. Cross-chapter pointer whose target carries the canonical tag 089J in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 0DES: sources/stacks-chow/tags.txt, source lines 16812–16812. Cross-chapter pointer whose target carries the canonical tag 0DES in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 0DDW: sources/stacks-chow/tags.txt, source lines 16781–16781. Cross-chapter pointer whose target carries the canonical tag 0DDW in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Lemma 0.1. *Let S be a scheme. Let $f : Y \to X$ be a surjective proper morphism of algebraic spaces over S. Let $\mathcal{F}$ be a sheaf on $X_{étale}$. Then $\mathcal{F} \to f_*f^{-1}\mathcal{F}$ is injective with image the equalizer of the two maps $f_*f^{-1}\mathcal{F} \to g_*g^{-1}\mathcal{F}$ where g is the structure morphism $g : Y \times_X Y \to X$.*

Proof. For any surjective morphism $f : Y \to X$ of algebraic spaces over S, the map $\mathcal{F} \to f_*f^{-1}\mathcal{F}$ is injective. Namely, if $\overline{x}$ is a geometric point of X, then we choose a geometric point $\overline{y}$ of Y lying over $\overline{x}$ and we consider

$$\mathcal{F}_{\overline{x}} \to (f_*f^{-1}\mathcal{F})_{\overline{x}} \to (f^{-1}\mathcal{F})_{\overline{y}} = \mathcal{F}_{\overline{x}}$$

See Properties of Spaces, Lemma 04K2 for the last equality.

The second statement is local on X in the étale topology, hence we may and do assume Y is an affine scheme.

Choose a surjective proper morphism $Z \to Y$ where Z is a scheme, see Cohomology of Spaces, Lemma 089J. The result for $Z \to X$ implies the result for $Y \to X$. Since $Z \to X$ is a surjective proper morphism of schemes and hence a ph covering (Topologies, Lemma 0DES) the result for $Z \to X$ follows from Étale Cohomology, Lemma 0DDW (in fact it is in some sense equivalent to this lemma). □

Lemma 0.2. *Let S be a scheme. Let X be an algebraic space over S. Let $\mathcal{F}$ be a sheaf on $X_{étale}$. Then $\pi_X^{-1}\mathcal{F}$ is given by the rule*

$$(\pi_X^{-1}\mathcal{F})(Y) = \Gamma(Y_{étale}, f_{small}^{-1}\mathcal{F})$$

for $f : Y \to X$ in $(Spaces/X)_{étale}$. Moreover, $\pi_Y^{-1}\mathcal{F}$ satisfies the sheaf condition with respect to smooth, syntomic, fppf, fpqc, and ph coverings.

Proof. Since pullback is transitive and $f_{small} = \pi_X \circ i_f$ (see above) we see that $i_f^{-1}\pi_X^{-1}\mathcal{F} = f_{small}^{-1}\mathcal{F}$. This shows that π_X^{-1} has the description given in the lemma.

To prove that $\pi_X^{-1}\mathcal{F}$ is a sheaf for the ph topology it suffices by Topologies on Spaces, Lemma 0DFM to show that for a surjective proper morphism $V \to U$ of algebraic spaces over X we have $(\pi_X^{-1}\mathcal{F})(U)$ is the equalizer of the two maps $(\pi_X^{-1}\mathcal{F})(V) \to (\pi_X^{-1}\mathcal{F})(V \times_U V)$. This we have seen in Lemma 0.1.

The case of smooth, syntomic, fppf coverings follows from the case of ph coverings by Topologies on Spaces, Lemma 0DFH.

Let $\mathcal{U} = \{U_i \to U\}_{i \in I}$ be an fpqc covering of algebraic spaces over X. Let $s_i \in (\pi_X^{-1}\mathcal{F})(U_i)$ be sections which agree over $U_i \times_U U_j$. We have to prove there exists a unique $s \in (\pi_X^{-1}\mathcal{F})(U)$ restricting to s_i over U_i. Case I: U and U_i are schemes. This case follows from Étale Cohomology, Lemma 09XN. Case II: U is a scheme. Here we choose surjective étale morphisms $T_i \to U_i$ where T_i is a scheme. Then $\mathcal{T} = \{T_i \to U\}$ is an fpqc covering by schemes and by case I the result holds for $\mathcal{T}$. We omit the verification that this implies the result for $\mathcal{U}$. Case III: general case. Let $W \to U$ be a surjective étale morphism, where W is a scheme. Then $\mathcal{W} = \{U_i \times_U W \to W\}$ is an fpqc covering (by algebraic spaces) of the scheme W. By case II the result hold for $\mathcal{W}$. We omit the verification that this implies the result for $\mathcal{U}$. □